

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
Hello World
Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
// You are using Java
import java.util.*;
```

```
class MessageAnalyzer {
    private TreeMap<Character, Integer> freqMap;
```

```
    public MessageAnalyzer() {
        freqMap = new TreeMap<>();
    }
```

```
    public void analyzeMessage(List<String> lines) {
```

```

        for (String line : lines) {
            for (char ch : line.toCharArray()) {
                if (Character.isLetter(ch)) { // ignore spaces and non-letters
                    freqMap.put(ch, freqMap.getDefault(ch, 0) + 1);
                }
            }
        }
    }

    public void printFrequencies() {
        System.out.println("Character Frequency:");
        for (Map.Entry<Character, Integer> entry : freqMap.entrySet()) {
            System.out.println(entry.getKey() + ": " + entry.getValue());
        }
    }

    public class Main {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);
            int n = Integer.parseInt(sc.nextLine());

            List<String> lines = new ArrayList<>();
            for (int i = 0; i < n; i++) {
                lines.add(sc.nextLine());
            }

            MessageAnalyzer analyzer = new MessageAnalyzer();
            analyzer.analyzeMessage(lines);
            analyzer.printFrequencies();
        }
    }

```

Status : Correct

Marks : 10/10